

Chapter 7 | Exponential Functions and Compound Interest

Algebra II · Unit 3: Exponential and Logarithmic Models · Grade 10

7.1 Linear Growth and Exponential Growth

A linear function adds the same amount in every time period. An exponential function multiplies by the same factor in every time period. This difference seems small at the start and becomes enormous over long spans, which is why exponential models are used for anything that grows on its own output: populations, infections, and money left to accumulate.

A savings account is the cleanest example. Interest earned in one year is added to the balance, and the next year interest is earned on the larger balance. Each year the growth is larger than the year before, even though the rate never changes.

7.2 The Compound Interest Formula

For a principal amount P invested at an annual rate r , compounded n times per year for t years, the final amount A is given by:

$$A = P (1 + r/n)^{nt}$$

The rate r is written as a decimal, so five percent is entered as 0.05. The exponent nt counts the total number of compounding periods over the whole investment.

7.3 Annual Compounding

When interest is compounded once per year, n equals 1 and the formula simplifies to the form used throughout this chapter:

$$A = P (1 + r)^t$$

This is the version to reach for in problems that specify annual compounding or that do not specify a compounding frequency at all.

7.4 How the Two Variables Behave

Two quantities control the outcome: the rate r and the time t . They do not act the same way, and recognizing the difference is the main goal of this section.

The rate sits inside the base. Changing it changes the multiplier applied in every single period, so its effect compounds on itself. Raising a rate from 4 percent to 8 percent does far more than double the final amount over a long span.

Time sits in the exponent. Each additional year multiplies the balance by the base one more time. Early years add little in absolute terms because the balance is small. Later years add a great deal, because the same percentage is applied to a much larger number.

A useful estimate is the rule of 72. Dividing 72 by the interest rate as a percentage gives the approximate number of years required for the balance to double. At 6 percent, a balance doubles in roughly 12 years.

7.5 Worked Example

A student invests 1,000 dollars at an annual rate of 5 percent, compounded annually. Find the balance after 10 years and after 30 years.

$$\text{After 10 years: } A = 1000 (1.05)^{10} = 1,628.89$$

$$\text{After 30 years: } A = 1000 (1.05)^{30} = 4,321.94$$

The first ten years added about 629 dollars. The last ten years of the thirty year span added roughly 1,665 dollars. The rate never changed. Only the size of the balance being multiplied changed.

7.6 Reaching a Target

Many problems ask the reverse question. Given a goal amount, what combination of rate and time will reach it? Because two variables appear, there is generally not one answer but a family of answers.

Suppose the goal is to turn 1,000 dollars into 3,000 dollars. A rate of 8 percent reaches that goal in about 14 years. A rate of 4 percent reaches the same goal in about 28 years. A rate of 11 percent reaches it in about 11 years. Each pairing satisfies the equation.

This is the practical trade off behind most saving decisions. A higher return shortens the wait. A longer horizon lowers the return required.

7.7 Exponential Decay

When a quantity decreases by a fixed percentage each period, the same formula applies with a negative rate. A car losing 15 percent of its value each year is modelled by $A = P(1 - 0.15)^t$. The curve falls steeply at first and then flattens, approaching but never reaching zero.

Practice

1. Find the balance on 2,500 dollars invested at 6 percent compounded annually for 18 years.
2. How many years does 1,000 dollars at 7 percent take to exceed 2,000 dollars?
3. Find two different pairs of rate and time that turn 500 dollars into 2,000 dollars, and explain why more than one pair works.
4. Compare 5,000 dollars at 3 percent for 40 years with 5,000 dollars at 6 percent for 20 years. Which ends higher, and why is the answer not obvious from the numbers alone?